



MG-1S各模块名称与功能原理介绍

Introduction to the MG-1S Modules and Features

飞行器正面

Front Side of the MG-1S



MG-1S 主控

MG-1S Main Controller



MG-1S主控主要协调飞行器各个功能模块的正常工作。

接线顺序：非常重要

M1-M8：连接M1-M8电调线；

M1357电调为白灰色；M2468电调为黄棕色；M1和M8电调线在同个主臂上；

CAN1：接GPS←气压计←中心板；

CAN2：接断电续航←中心板电源线；

iESC:接断电续航信号线←四个电调馈线；

RF：接接收机←中心板上的对频板；

LED:接LED模块；

The MG-1S main controller mainly coordinates all functional modules of the MG-1S.

Cable connection sequences shown below are important.

M1 to M8: Connected with M1 to M8 ESC cables.

The M1357 ESC is white and grey while the M2468 ESC is yellow and brown. M1 and M8 ESC cables are on one main arm.

CAN1: Connected with GPS ← Barometer ← Central board

CAN2: Connected with reserved battery module ← Central board power cable

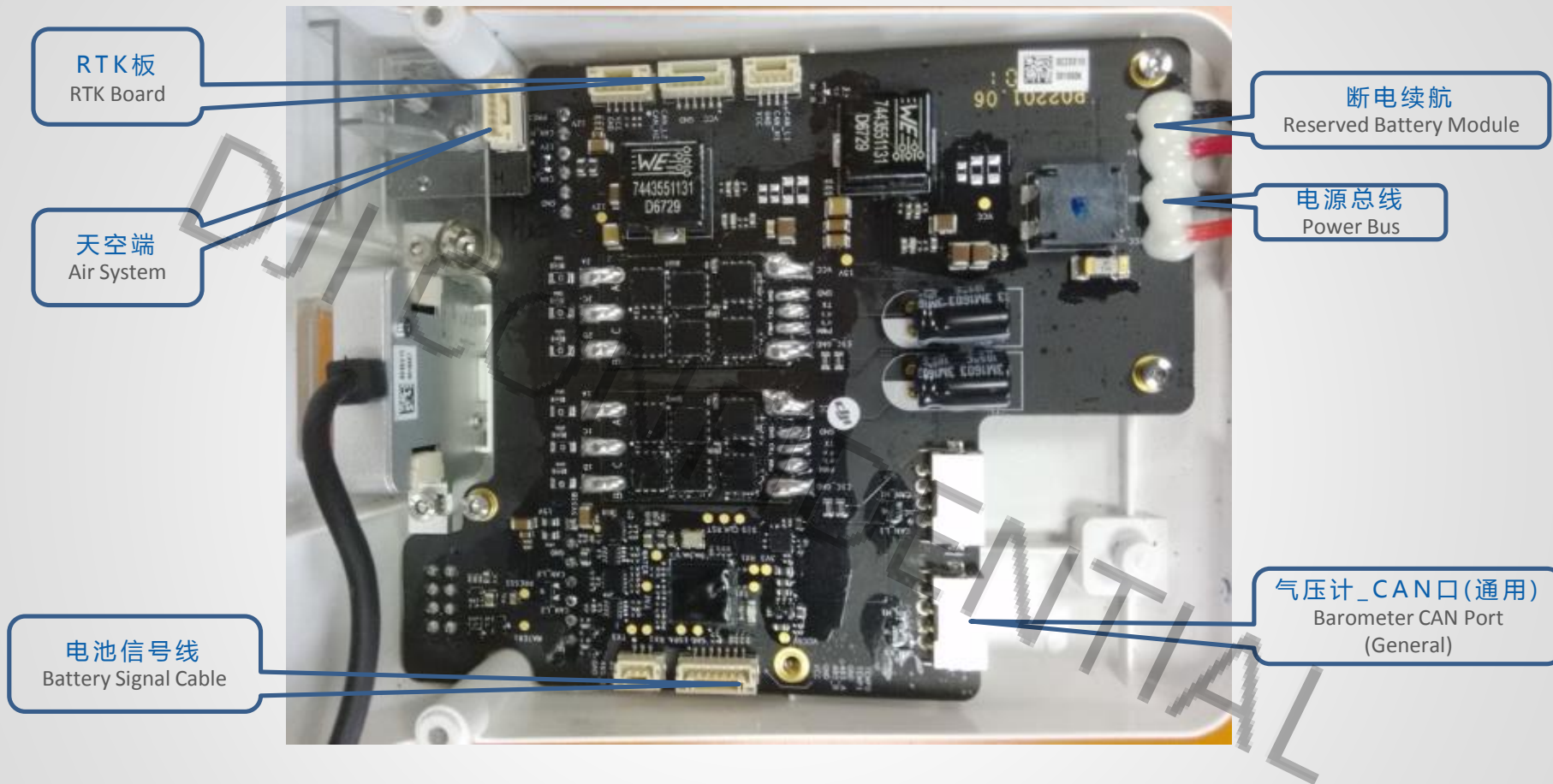
iESC: Connected with reserved battery module signal cable ← Four ESC feeders

RF: Connected with the receiver ← Link board on the central board

LED: Connected with the LED module

中心板+对频板

Central Board and Link Board



中心板主要是电源转换模块，将12S的电源转换成各个模块所需要的电压；对频板与天空端连接，是天空端的对频按键模块；

The central board serves as a power conversion module which converts a 12S power supply to the voltage required by each module.

Link board is the link button module and connected with the air system.

RTK Board



GPS



GPS：全球定位系统

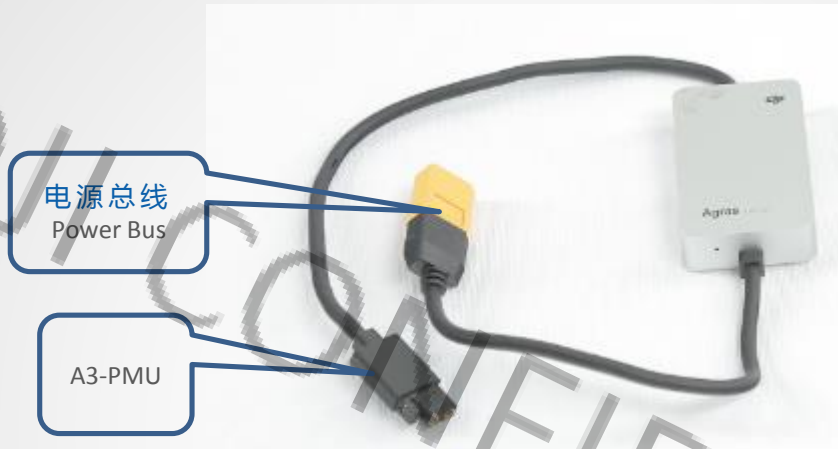
MG-1S通过GPS模块与指南针模块进行定高定位和方向识别，实现飞行器控制功能；GPS分别连接右脚架前后的两个指南针，切记连接不可混淆；GPS线材上的CAN与气压计相连。

Global Positioning System (GPS)

The MG-1S utilizes the GPS and compass modules to measure the height, position, identify directions.

The GPS module needs to be connected with compass in the front and rear landing gear, respectively. They must be connected correctly. The CAN port of the GPS cable is connected with the barometer.

PMU



PMU：电源管理单元

PMU模块主要为管控提供稳定电源。

PMU电源总线接智能电池模组，输出端接A3主控的PMU接口；

Power Management Unit (PMU)

PMU mainly supplies power to the main controller.

The power bus of the PUM is connected with the Intelligent Flight Battery modules. The output end is connected with the PMU port on the A3 main controller.

LED



LED指示灯用于飞行过程中指示飞控系统的状态。USB口主要用于连接PC端 DJI Assistant2进行参数调节与主控固件升级。LED连接主控A3的LED端口。

The LED indicator shows the status of the flight controller during flight. The USB port is used to connect with a computer installed with DJI Assistant 2 for setting parameters and updating the main controller firmware. LED is connected with the LED port of the A3 main controller.

天空端 Air System

Lightbridge 2

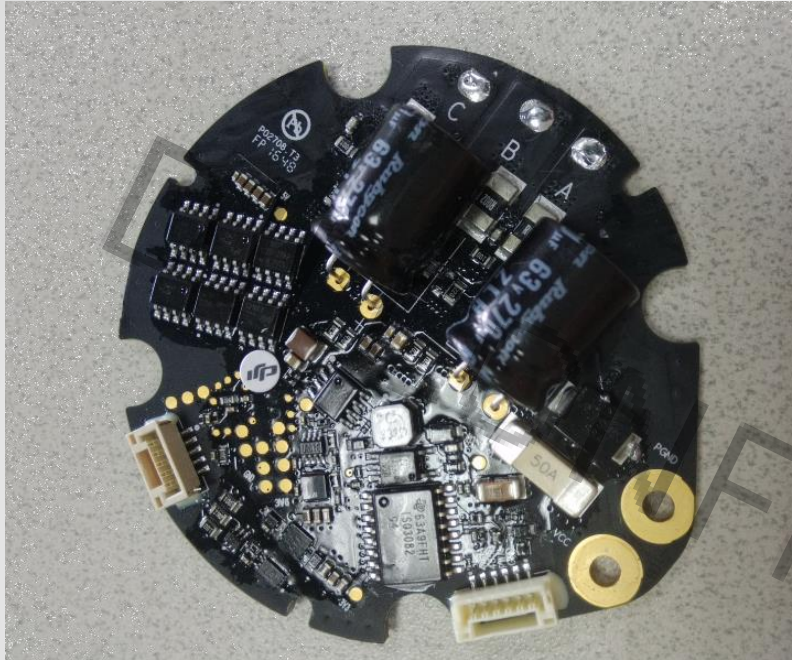


天空端模块主要负责与遥控器进行通信；与A3主控系列的天空端不一样；天空端与A3主控的RF端口和对频板连接。1号天线最短，2和3天线一样，4号天线最长。1号天线安装在机头，2号天线安装在下壳右边，3号天线安装在机尾，4号天线安装下壳左边。

The air system module is mainly used for communication with the remote controller. It is different from the air system of the A3 main controller series. The air system is connected with the RF port and link board of the A3 main controller. Antenna 1 is the shortest while antenna 4 is the longest. Antenna 2 and antenna 3 are of the same length.

电机和电调

Motor and ESC



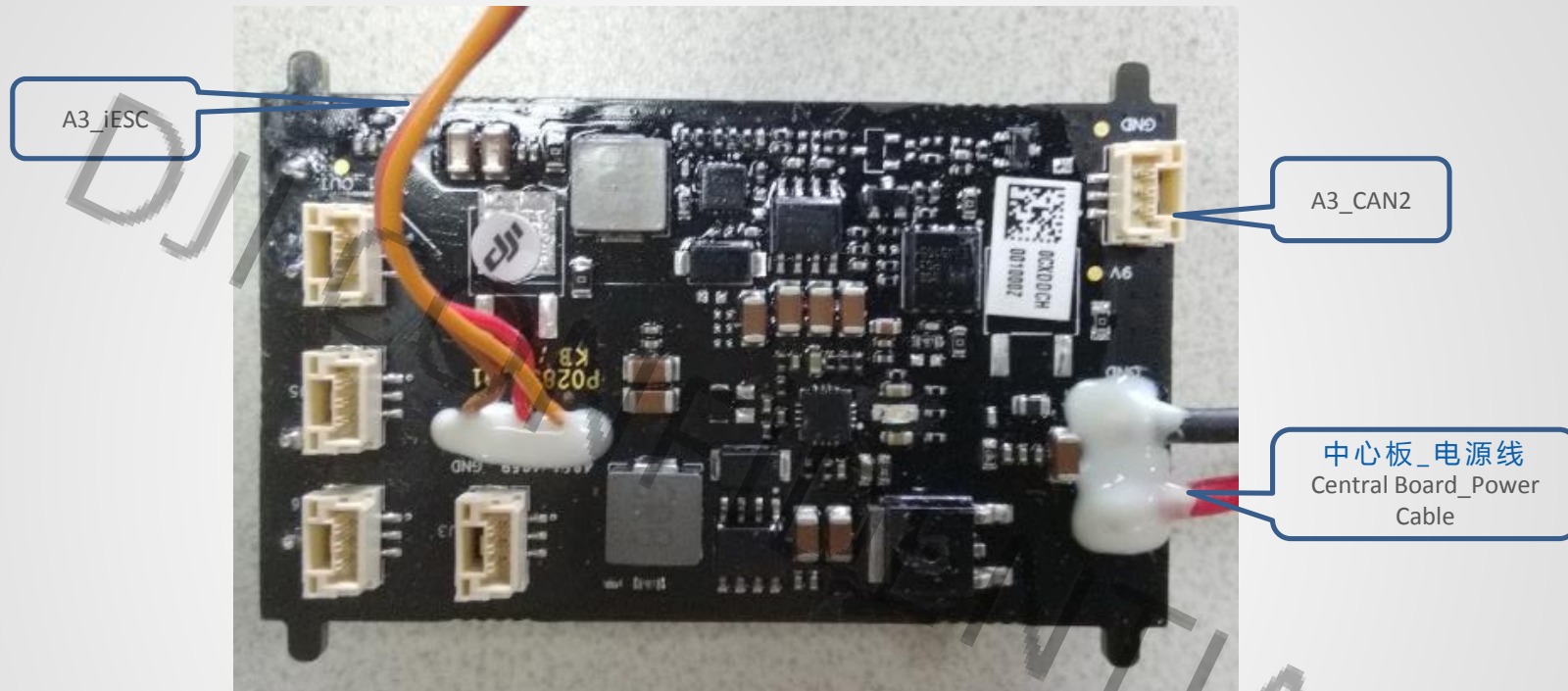
M1、M3、M5、M7电机与电调的接法：A、B、C分别接电机线红、黑、蓝；
M2、M4、M6、M8电机与电调的接法：A、B、C分别接电机线红、蓝、黑；
八个电调一样，八个电机一样。

When connecting motor 1, motor 3, motor 5 and motor 7 to the ESCs, ensure that A is connected to the red motor cable, B is connected to the black motor cable and C is connected to the blue motor cable.

When connecting motor 2, motor 4, motor 6 and motor 8 to the ESCs, ensure that A is connected to the red motor cable, B is connected to the blue motor cable and C is connected to the black motor cable.

All the eight ESCs and motors are the same.

断电续航模块 Reserved Battery Module



断电续航模块是MG-1S的备用电源模块，主要是飞行器在更换电池时保持主控不断电，避免自动模式下记录的AB点数据丢失；（J3、J4、J5、J6四个接口接四个主臂的电调反馈线，无区别）

The MG-1S's reserved battery module supplies power to the main controller when there is no battery in the MG-1S, which prevents the data of point A and point B recorded in auto mode from losing. (J3、J4、J5、J6 are the ports for four ESC feedback lines from main arms, there are no difference between these ports)

雷达+液压计 Radar and Fluid Level Meter



前中后三个雷达主要分别检测飞行器与前方坡度、底下农作物、后方坡度的距离，反馈给主控调节，使飞行器在工作时提前预知周围环境是否有较大的坡度或落差。前雷达灯为蓝色，中间雷达灯为黄色，后雷达灯为绿色。

Three radars on the front, center and rear of the MG-1S measure the distance between the slope in front of the MG-1S and the MG-1S, crops and the MG-1S, the slope at the back of the MG-1S and the MG-1S, and report the data to the main controller, which allows the aircraft to know whether there is a steep slope or large height difference in the surroundings in advance during operation. The front radar light is red, the middle radar light is yellow, the back radar light is green.

液位计

Fluid Level Meter



液位计主要检测水箱剩余药量，并反馈给主控。

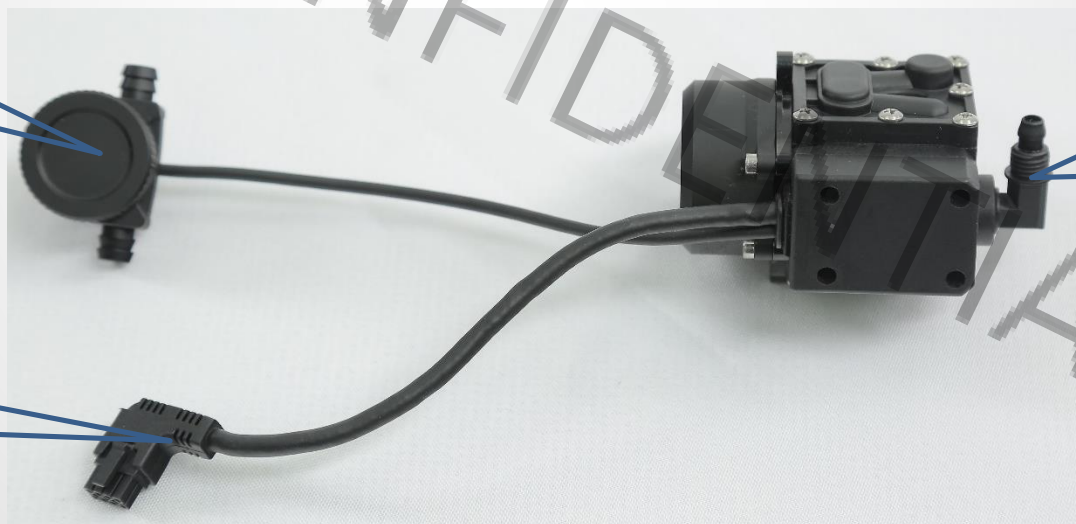
Fluid level meter measures the remaining liquid in the Spray Tank and reports this information to the main controller.

水泵 Delivery Pump



中心板_右
Right Central Board

M 4和 M 7 喷头
M4 Sprinkler and
M7 Sprinkler



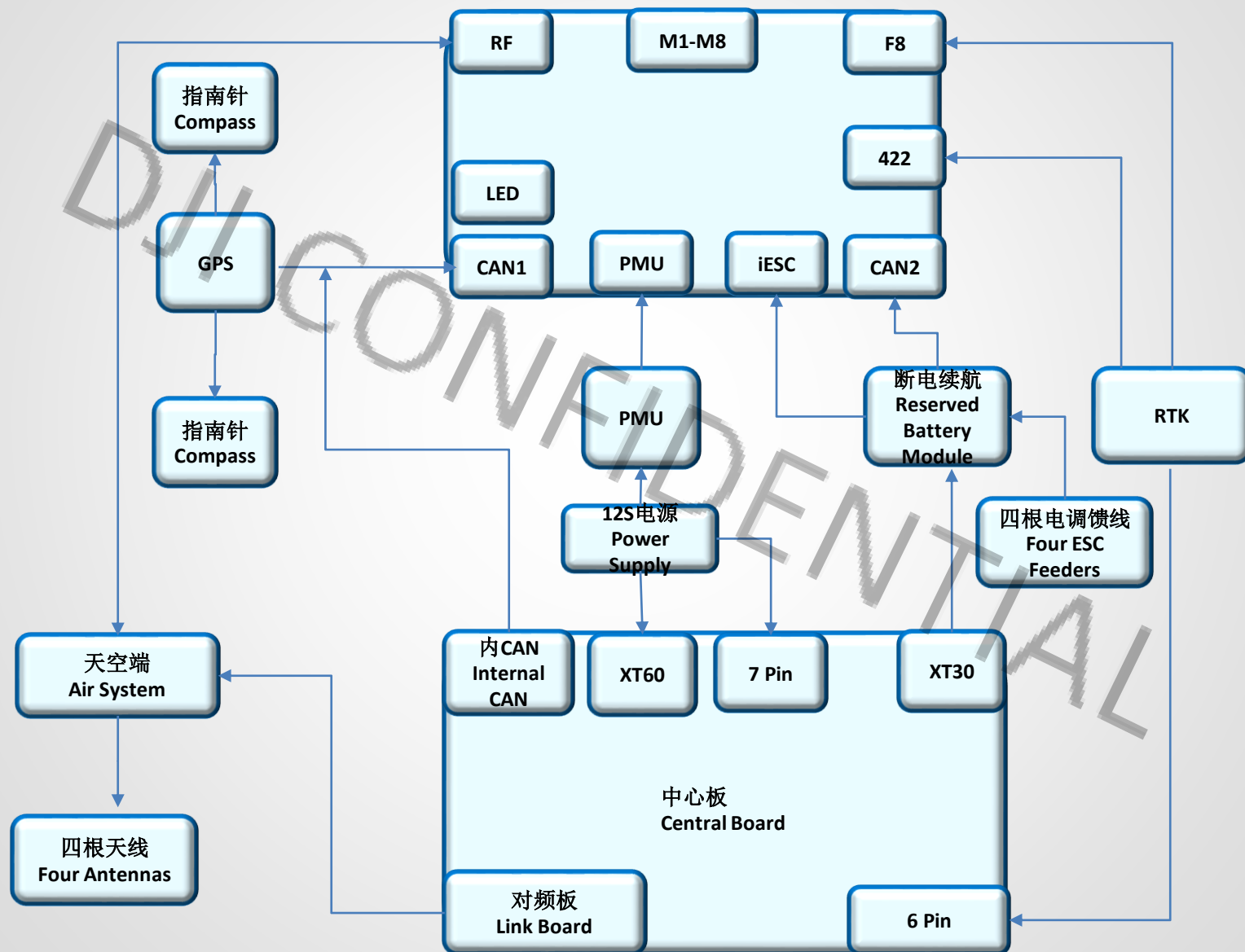
流量计
Flowmeter

M 3和 M 8 喷头
M3 Sprinkler and
M8 Sprinkler

中心板_左
Left Central Board

整机接线图

MG-1S Cable Connection



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**THANK
YOU**